

**A Practical Use Of AI For Church Sermon Analysis**

Author: Luke Flattery, PhD

Contact: [lf7127725@gmail.com](mailto:lf7127725@gmail.com)

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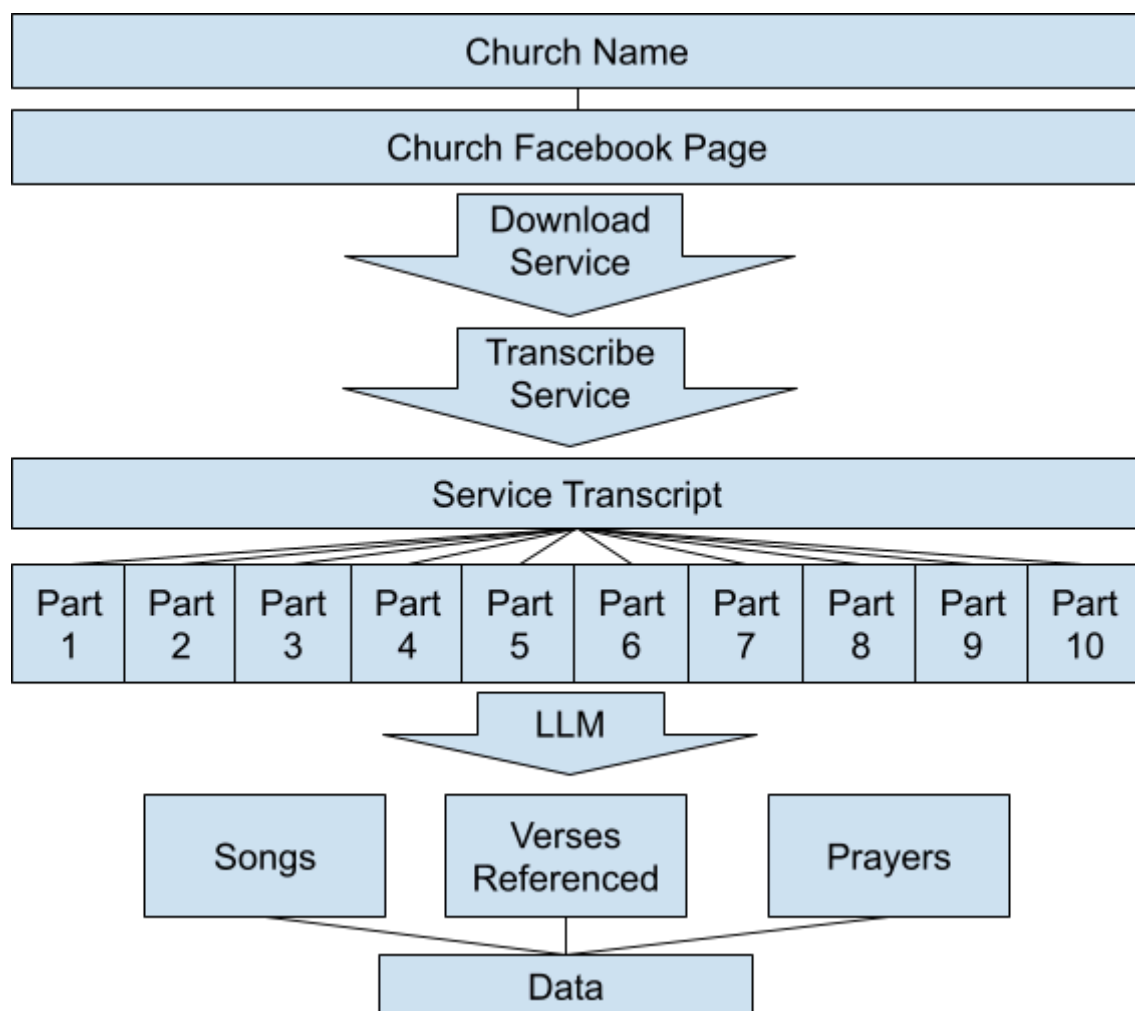
How can the church utilize the AI technology boom to glorify God? If there are 300 churches in a district averaging two services a week that would mean 31,200 services are conducted a year. How many of these services are consumed by any single person? How many worship leaders are capable of analyzing all worship sets that take place over the year in their church district to find trends in the number and diversity of songs by month or church? How many pastors know the popularity of books and verses being preached in their district and can provide transcripts from all churches of all sermons preached referencing a certain verse? How many church districts can list and categorize all prayers being prayed across all churches in the district?

The data needed to answer these questions are easily obtained through modern AI techniques and the following will provide an exact roadmap for answering these questions. The goal is that a district could give this project to an intern or a volunteer with basic programming knowledge to replicate the results of this paper for a chosen group of churches with minimal resources. The point isn't the comparison of God's calling on church leaders or the generation of AI trash. Why can't God use AI to support the training of pastors and equipping of new believers?

The only requirement to start this project is a list of the churches you want to study, and a link to their Facebook page. The only real cost to the project is the LLM usage (Chat GPT 4o), which came to around \$50. However, a cheaper open source model (such as Llama 4 or Deepseek) could definitely produce the same result for a fraction of the cost. If done again, this project could be done with a budget of less than \$2 for LLM API usage. This paper uses data from about 170 church services spread across 13 churches to demonstrate what is possible.

### Methodology

The methodology utilizes an automated pipeline to collect, download, and transcribe live-streamed video sermons from the Facebook page of a church. The overall aim is to transform publicly available video content into text so it can be further analyzed for themes, frequency of messages, or other content-based insights. The process combines several tools—web automation for data collection, video downloading, and state-of-the-art speech-to-text technology for transcription. The methodology is designed to be scalable and minimally dependent on manual input, requiring only the Facebook page link of the church. The chart below shows the flow of the project.



The steps needed to complete this project are outlined below with the complete code used available for full details.

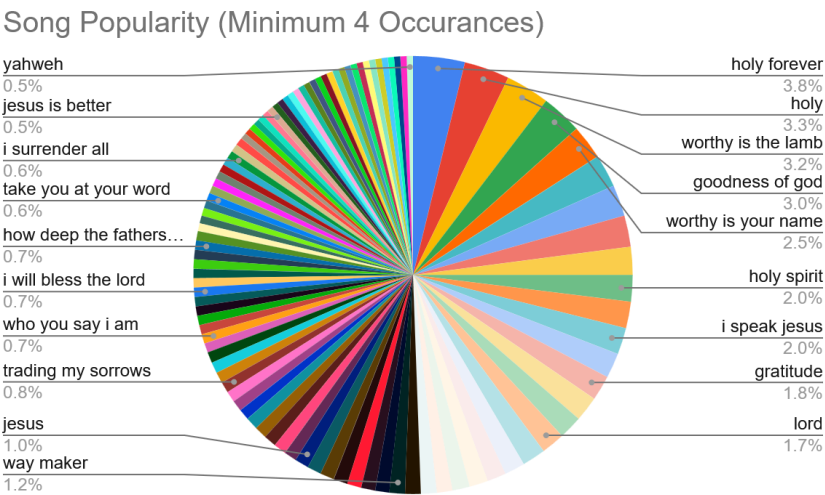
1. **Identify Services** - The Facebook page for a church can be web-scraped to find all available live videos in order of publishing. The system is designed to identify services within a specified date range. There is nothing special about Facebook pages here, the same could be done for YouTube or other sites.
2. **Download and Transcribe** - Using the Python library yt-dlp, the audio from the videos is downloaded. Each video is transcribed using an implementation of OpenAI's Whisper. Once videos are transcribed the audio files can be deleted.
3. **Prepare Data** - Each transcript is split into 10 slightly overlapping parts. This is a core component of this method. By doing this you can mark each part of a service as worship, prayer, sermon, etc. Keeping the service in parts also helps the LLM process the data.
4. **Apply LLM** - Using a system of 30 calls to an LLM API, the transcript is converted to data. This paper focuses on songs present, verses referenced, and prayers prayed. This part of the process can easily be changed to fit different use cases.
5. **Analyze Results** - There isn't anything novel about the analysis of the data. If this paper is significant, it is because the data itself hasn't been previously collected. The data is the key takeaway, the results are merely an initial take on the underlying value.

Results

The data collected for this study comes from 17 of the largest churches in the Assemblies of God according to an outdated Wikipedia page. The services collected mostly span from January to March 2025 with a few exceptions. It would not be difficult to apply the same code to 1000 churches or use a different time frame. The data collected included data on worship songs, verses referenced, and prayers.

Worship Songs

The top songs across all churches can be identified.



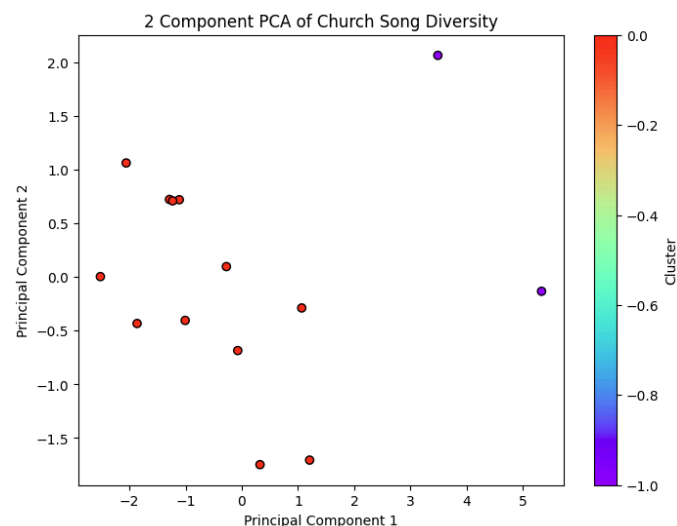
The same could be done for specific timeframes.

| Song Popularity By Month (Number of Occurrences) |    |                     |    |                     |    |
|--------------------------------------------------|----|---------------------|----|---------------------|----|
| Mar 25                                           |    | Feb 25              |    | Jan 25              |    |
| all hail king jesus                              | 13 | holy                | 15 | holy forever        | 21 |
| worthy is the lamb                               | 10 | great are you lord  | 13 | goodness of god     | 16 |
| gratitude                                        | 7  | worthy is the lamb  | 12 | hallelujah          | 15 |
| same god                                         | 5  | worthy is your name | 11 | holy                | 14 |
| great is your faithfulness                       | 5  | who else is worthy  | 11 | worthy is your name | 11 |

Stats can be run to compare churches on the number of songs run per service. Some of these could be songs played in the background before service, but the system could be improved to remove this.

| Count | Mean        | Std Dev      | Min | Max | Church Name                |
|-------|-------------|--------------|-----|-----|----------------------------|
| 52    | 2.230769231 | 2.044748267  | 1   | 13  | City_First_Church_Rockford |
| 82    | 1.390243902 | 0.8854948754 | 1   | 6   | Faith_Assembly_Orlando     |
| 118   | 1.457627119 | 0.8437205739 | 1   | 5   | First_Assembly_Ft_Myers    |
| 62    | 1.290322581 | 0.6372866862 | 1   | 4   | First_Assembly_Griffin_GA  |
| 54    | 1.555555556 | 0.8392869091 | 1   | 4   | First_Assembly_NLR         |
| 52    | 1.576923077 | 0.893245983  | 1   | 5   | Grand_Rapids_First         |
| 38    | 1.315789474 | 0.5253191132 | 1   | 3   | Life_Center_Tacoma_WA      |
| 89    | 1.865168539 | 1.791336545  | 1   | 13  | Life_Church_Midlothian     |
| 188   | 1.680851064 | 1.293445144  | 1   | 8   | New_Life_Covenant_Chicago  |
| 178   | 1.921348315 | 1.761000737  | 1   | 10  | Oaks_Church                |
| 82    | 1.719512195 | 1.240015833  | 1   | 7   | River_Valley_Church        |
| 170   | 2.529411765 | 2.583920398  | 1   | 17  | Timberline_Church          |
| 128   | 1.7109375   | 1.236914779  | 1   | 7   | Trinity_Church_Cedar_Hill  |
| 56    | 1.625       | 0.7760623218 | 1   | 4   | Westover_Hills_San_Antonio |

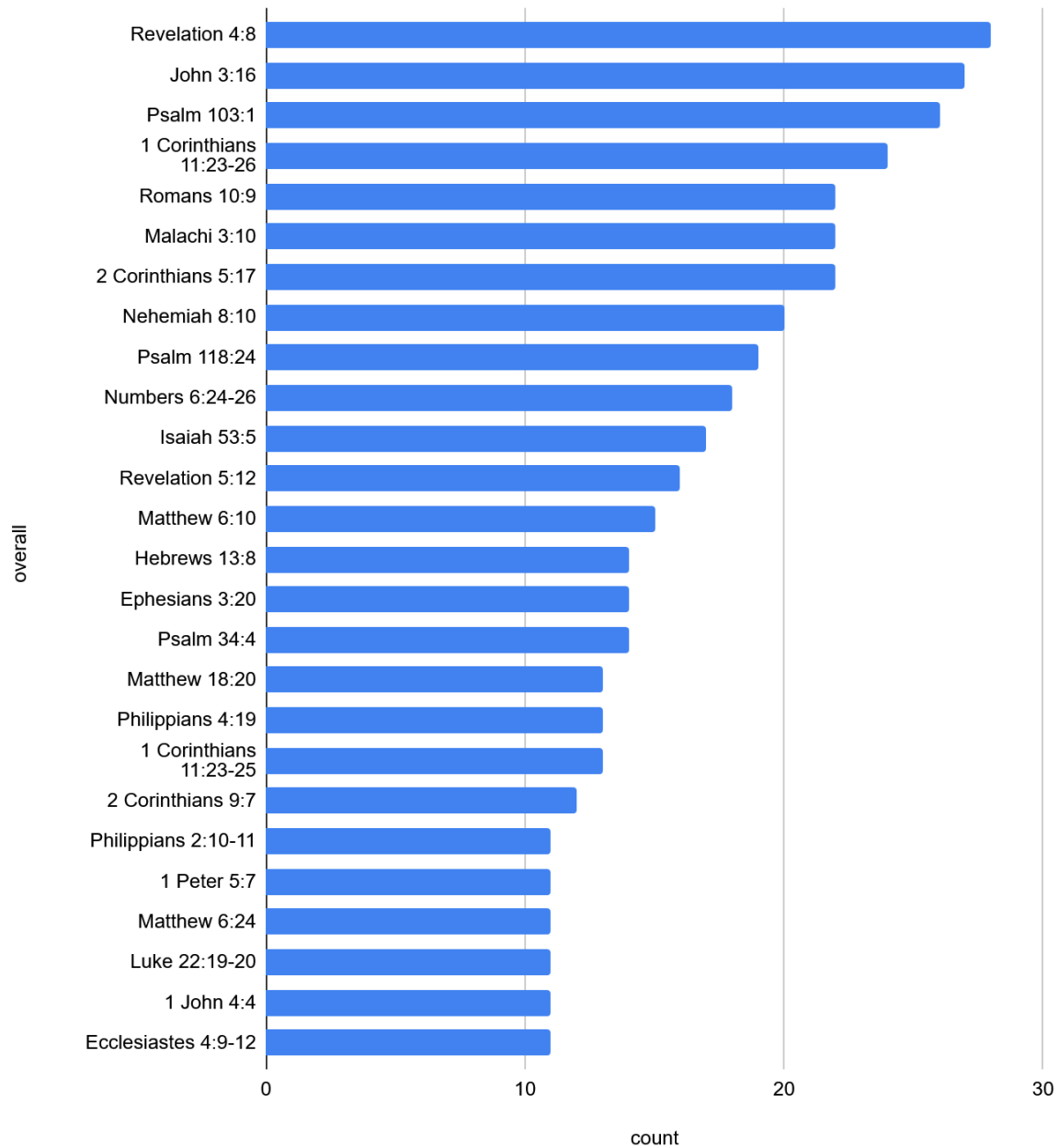
A machine learning approach could take the same data and detect outliers. This approach works well even when comparing thousands of objects. Each dot is a specific church in this case.



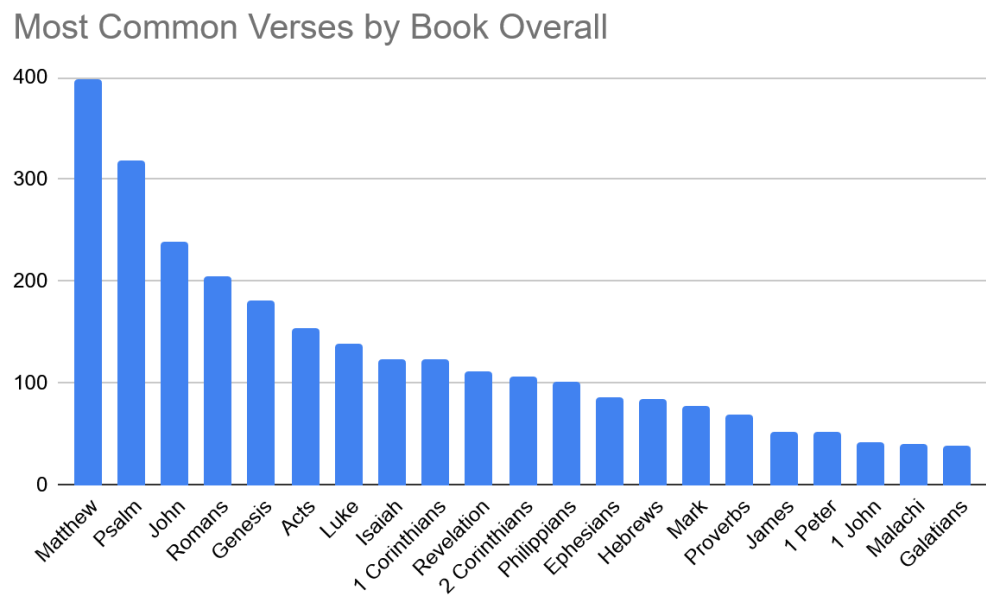
## Verse Analysis

What are the top verses being quoted in this sample of services?

### Top Verses Overall (Minimum 10 Occurances)

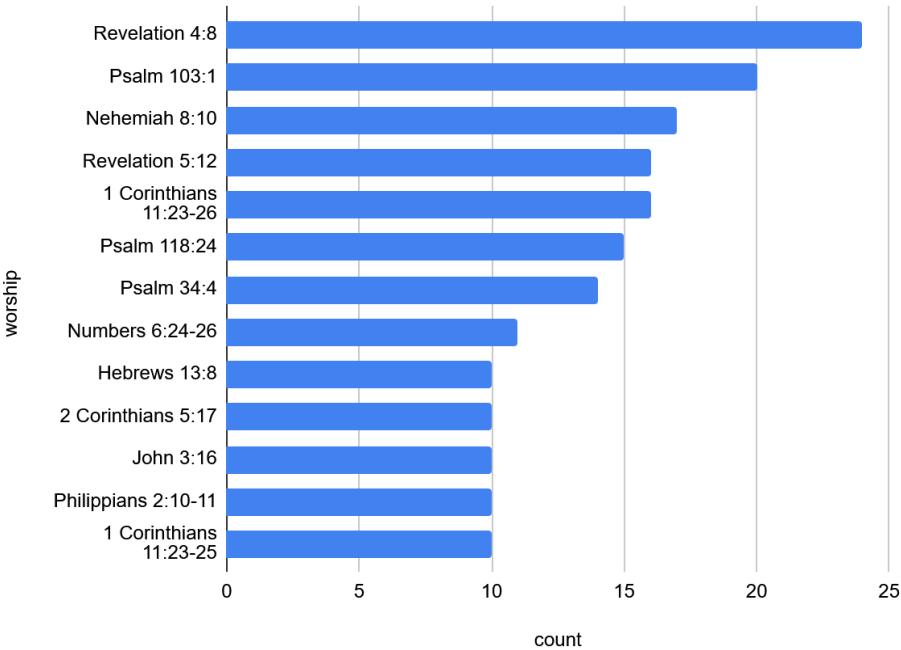


What about books of the Bible being referenced throughout entire services?



The system used can mark each part of a transcript as having songs. What are the top verses quoted in parts of the transcript with songs (including the songs themselves and transitions)?

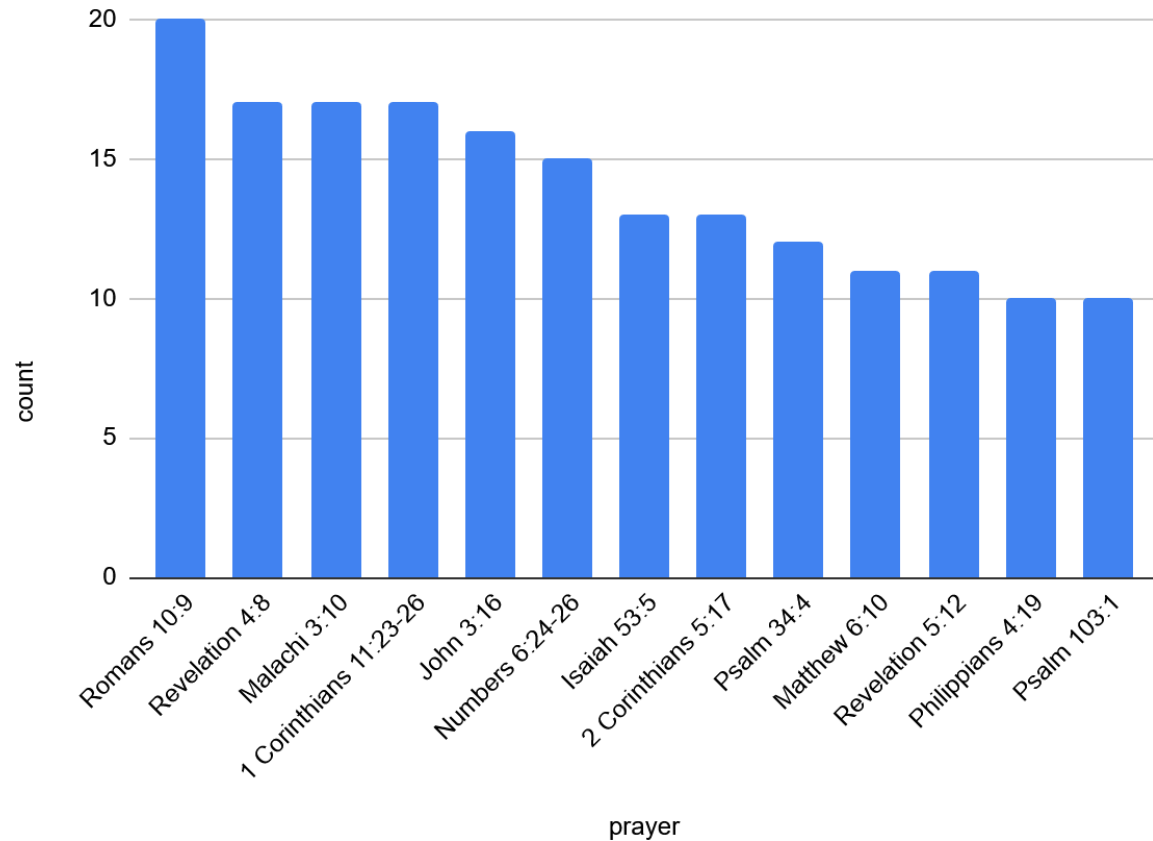
Top Verses in Worship (Minimum 10 Occurances)





What about the most common verses referenced during prayers?

### Top Verses Quoted During Prayers (Minimum 10 Occurances)

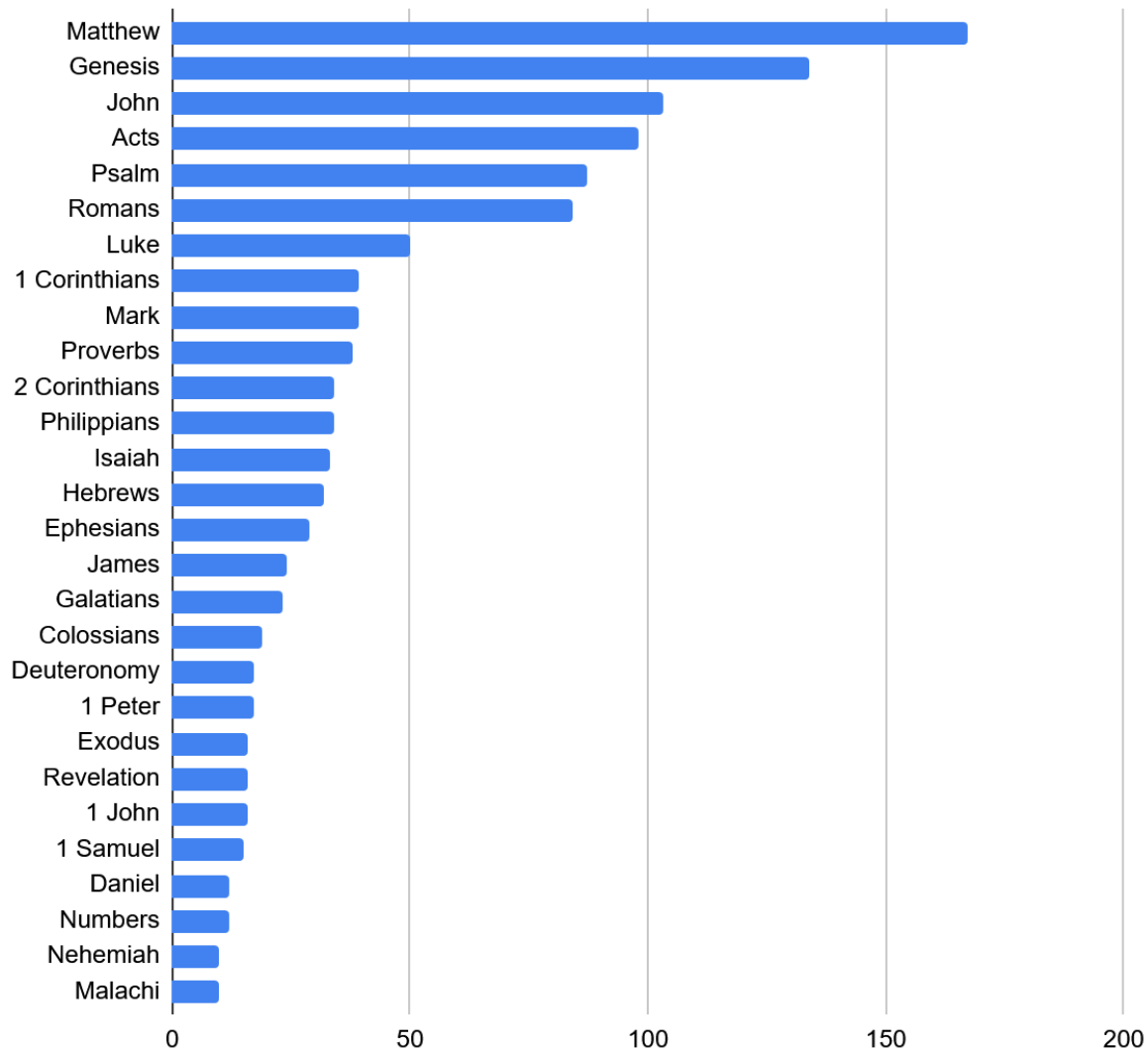


The top verses during sermons don't have as interesting of a graph. All of the following occur 5-10 times.

|              |                |                    |                     |               |
|--------------|----------------|--------------------|---------------------|---------------|
| Matthew 6:24 | Acts 1:8       | Hebrews 12:2       | Romans 6:23         | Matthew 25:40 |
| Romans 12:1  | Acts 2:42-47   | 1 Corinthians 5:11 | Ecclesiastes 4:9-12 | Romans 15:13  |
| John 3:16    | Proverbs 18:24 | Acts 20:35         | Matthew 4:18-22     | Genesis 2:18  |

What are the top books during sermons?

### Top Books Exclusive (Sermon)



### Topic Modeling Prayers

What if we apply topic modeling to the prayers identified in the data? Topic modelling is a machine learning technique that will categorize prayers and explain why each prayer falls into a certain category. Further analysis could see if topics such as prayers for healing line up with certain messages being preached, or if prayers related to offering or tithe happen more often in

specific months. Instead of prayers, topic modelling could have been done on offering segments, calls to salvation, or any other common component of services that could be added to the dataset.

Of the 1488 prayers in the data collected, below are the major topics.

| Number of Prayers | Topic Description                       |
|-------------------|-----------------------------------------|
| 199               | Thanks                                  |
| 77                | Service closing words                   |
| 63                | Asking for specific needs in Jesus name |
| 55                | Salvation                               |
| 40                | Tithe and giving prayers                |
| 37                | Healing                                 |
| 37                | Specific Church Events                  |

**Conclusion**

The results shown in this paper will be useful for the analysis of services as a group. No single service can be judged without observing the actual service. This tool opens the door to studies pertaining to the needs and initiatives taking place in a group of churches. What insights into a collection of services would prove beneficial? This paper contains examples of numerical metrics, time analysis, and topic categorization. A couple modern data analysis techniques that scale to a dataset thousands of times bigger than the sample used have been demonstrated. The truly valuable data is ready to be drawn out by groups that have a strong understanding of the anatomy of services and common church practices.

This is only the beginning. The output of this process is the transcript from services and these transcripts could be a tremendous tool to be used by the church. A catalog of messages could be searched to find what verses are typically preached about a certain topic. Research could be done on how a certain type of prayer is being done in other churches. Congregations could be directed to sermons that answer their questions in video format, already prepared and available online for them to consume. Training for running all aspects of a service could be improved with big data. Finally, take note that none of this requires the technology of tomorrow or even a large amount of resources.

## References

Largest Churches in the AG List

[https://en.wikipedia.org/wiki/List\\_of\\_megachurches\\_affiliated\\_with\\_the\\_Assemblies\\_of\\_God](https://en.wikipedia.org/wiki/List_of_megachurches_affiliated_with_the_Assemblies_of_God)

BERTopic

<https://maartengr.github.io/BERTopic/index.html>

Sklearn PCA

<https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html>

Chat gpt 4o API

<https://openai.com/index/chatgpt/>

## Churches Used

Timberline Church, Fort Collins, Colorado

[https://www.facebook.com/timberlinechurch/live\\_videos](https://www.facebook.com/timberlinechurch/live_videos)

First Assembly of God, North Little Rock, Arkansas

[https://www.facebook.com/FirstNLR/live\\_videos](https://www.facebook.com/FirstNLR/live_videos)

New Life Covenant Assemblies of God, Chicago, Illinois

[https://www.facebook.com/MyNewLifeOrg/live\\_videos](https://www.facebook.com/MyNewLifeOrg/live_videos)

River Valley Church, Apple Valley, Minnesota

[https://www.facebook.com/rivervalleychurch/live\\_videos](https://www.facebook.com/rivervalleychurch/live_videos)

Trinity Church, Cedar Hill, Texas

[https://www.facebook.com/trinitychurchcedarhill/live\\_videos](https://www.facebook.com/trinitychurchcedarhill/live_videos)

First Assembly of God, Fort Myers, Florida

[https://www.facebook.com/FirstAssemblyofGod/live\\_videos](https://www.facebook.com/FirstAssemblyofGod/live_videos)

First Assembly of God Life Center, Tacoma, Washington

[https://www.facebook.com/LifeCenterCentral/live\\_videos](https://www.facebook.com/LifeCenterCentral/live_videos)

City First Church, Rockford, Illinois

[https://www.facebook.com/cityfirstchurch/live\\_videos](https://www.facebook.com/cityfirstchurch/live_videos)

Westover Hills Assembly of God, San Antonio, Texas

[https://www.facebook.com/westoverhills/live\\_videos](https://www.facebook.com/westoverhills/live_videos)

Faith Assembly of God, Orlando, Florida

[https://www.facebook.com/faithorl/live\\_videos](https://www.facebook.com/faithorl/live_videos)

First Assembly of God, Griffin, Georgia

[https://www.facebook.com/Griffinfirst/live\\_videos](https://www.facebook.com/Griffinfirst/live_videos)

First Assembly of God, Grand Rapids, Michigan

[https://www.facebook.com/GrandRapidsFirst/live\\_videos](https://www.facebook.com/GrandRapidsFirst/live_videos)

Life Church, Midlothian, Texas

[https://www.facebook.com/lcmidlo/live\\_videos](https://www.facebook.com/lcmidlo/live_videos)

Oaks Church, Red Oak, Texas

[https://www.facebook.com/oakschurctx/live\\_videos](https://www.facebook.com/oakschurctx/live_videos)